

Molecular Tumor Boards in Cancer Care: Ripe for Machine Learning?

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Navigating the Complexity of Cancer Treatment

Cancer treatment is complex. Understanding the intricate molecular pathways altered in cancer is crucial for effective therapy. This task requires navigating the multifaceted landscape of genetic aberrations, tumor evolution, and the challenges of selecting the right targeted therapy at the right time for the patient.



Overview of Molecular Tumor Board (MTB)

Definition:

- An interdisciplinary group focused on reviewing molecular and genetic test results to inform personalized cancer treatment.



Key Functions:

- Reviewing complex cancer cases
- Recommending targeted therapies, clinical trials, and further genetic testing.

Why Were Molecular Tumor Boards Established?



Precision medicine and biomarker testing have transformed cancer care by identifying actionable alterations that can guide clinical care.



MTBs emerged to help clinicians interpret increasingly complex genetic data.



Barriers like limited clinician support and data complexity still hinder precision medicine uptake.

Why use MTBs?

- Determining the best treatment for the patient requires
 - An understanding of standard of care therapies
 - Knowledge of the altered molecular pathways
 - Outcomes data for targeted therapies
 - Clinical trials available
- This is complicated



Translating Biomarker Results into Individualized Recommendations

Start with the individual themselves:

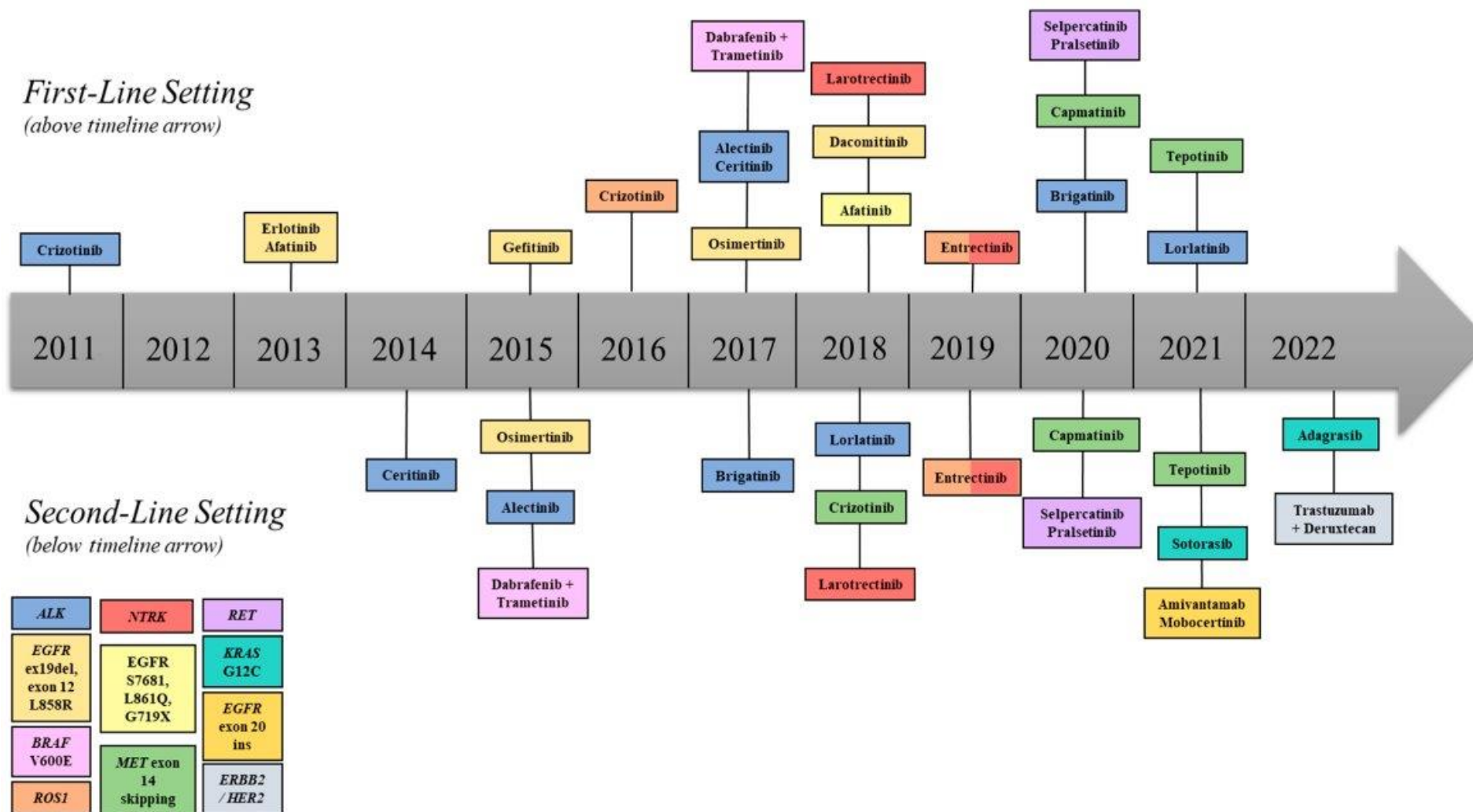
- Cancer diagnosis and stage
- Prior therapies
- Goals of care
- Other important comorbidities or considerations

Biomarker tests results have limited patient specific information

Goal is to translate the findings into realistic treatment options for each individual patient

- Summarize multiple genomic tests → tumors can change over time
- Consider current and future therapies
- Consider clinical trial options
- Consider if germline testing is indicated

Targeted Therapies in Non-Small Cell Lung Cancers



Disease Type	Approved Drugs
Bladder Cancer	Atezolizumab, Avelumab, Enfortumab vedotin-ejfv, Erdafitinib, Nivolumab, Nogapendekin alfa inbakicept-pmIn, Pembrolizumab, Sacituzumab govitecan-hziy
Brain Cancer	Belzutifan, Bevacizumab, Dabrafenib, Everolimus, Tovorafenib, Trametinib, Vorasidenib
Breast Cancer	Abemaciclib, Ado-trastuzumab emtansine, Alpelisib, Anastrozole, Capivasertib, Elacestrant dihydrochloride, Everolimus, Exemestane, Fam-trastuzumab deruxtecan-nxki, Fulvestrant, Goserelin acetate, Lapatinib ditosylate, Letrozole, Margetuximab-cmkb, Neratinib maleate, Olaparib, Palbociclib, Pembrolizumab, Pertuzumab, Ribociclib, Sacituzumab govitecan-hziy, Talazoparib tosylate, Tamoxifen citrate, Toremifene, Trastuzumab, Tucatinib
Cervical Cancer	Bevacizumab, Pembrolizumab, Tisotumab vedotin-tftv
Colorectal Cancer	Adagrasib, Bevacizumab, Cetuximab, Encorafenib, Fruquintinib, Ipilimumab, Nivolumab, Panitumumab, Pembrolizumab, Ramucirumab, Regorafenib, Tucatinib, Ziv-aflibercept
Dermatofibrosarcoma Protuberans	Imatinib mesylate
Endocrine/Neuroendocrine Tumors	Avelumab
Endometrial Cancer	Dostarlimab-gxly, Durvalumab, Lenvatinib mesylate, Pembrolizumab
Esophageal Cancer	Fam-trastuzumab deruxtecan-nxki, Ipilimumab, Nivolumab, Pembrolizumab, Ramucirumab, Tislelizumab-jsgr, Trastuzumab
Head and Neck Cancer	Cetuximab, Nivolumab, Pembrolizumab, Toripalimab-tpzi
Gastrointestinal Stromal Tumor	Avapritinib, Imatinib mesylate, Regorafenib, Ripretinib, Sunitinib malate
Giant Cell Tumor	Denosumab, Pexidartinib hydrochloride
Kidney Cancer	Avelumab, Axitinib, Belzutifan, Bevacizumab, Cabozantinib-s-malate, Everolimus, Ipilimumab, Lenvatinib mesylate, Nivolumab, Pazopanib hydrochloride, Pembrolizumab, Sorafenib tosylate, Sunitinib malate, Temsirolimus, Tivozanib hydrochloride
Leukemia	Acalabrutinib, Alemtuzumab, Asciminib hydrochloride, Avapritinib, Blinatumomab, Bosutinib, Brexucabtagene autoleucel, Dasatinib, Duvelisib, Enasidenib mesylate, Gemtuzumab ozogamicin, Gilteritinib fumarate, Glasdegib maleate, Ibrutinib, Idelalisib, Imatinib mesylate, Inotuzumab ozogamicin, Ivosidenib, Lisocabtagene maraleucel, Midostaurin, Nilotinib, Obinutuzumab, Ofatumumab, Olutasidenib, Pemigatinib, Pirtobrutinib, Ponatinib hydrochloride, Quizartinib dihydrochloride, Rituximab
Liver/Bile Duct Cancer	Atezolizumab, Bevacizumab, Cabozantinib-s-malate, Durvalumab, Futibatinib, Ipilimumab, Ivosidenib, Lenvatinib mesylate, Nivolumab, Pembrolizumab, Pemigatinib, Ramucirumab, Regorafenib, Sorafenib tosylate, Tremelimumab-actl
Lung Cancer	Adagrasib, Afatinib dimaleate, Alectinib, Amivantamab-vmjw, Atezolizumab, Bevacizumab, Binimetinib, Brigatinib, Capmatinib hydrochloride, Cemiplimab-rwlc, Ceritinib, Crizotinib, Dabrafenib mesylate, Dacomitinib, Durvalumab, Encorafenib, Entrectinib, Erlotinib hydrochloride, Fam-trastuzumab deruxtecan-nxki, Gefitinib, Ipilimumab, Lazertinib mesylate hydrate, Lorlatinib, Necitumumab, Nivolumab, Osimertinib mesylate, Pembrolizumab

Current FDA Approved Targeted Therapies

List of Some of the Non-FDA Approved drugs in Clinical Trials

Disease Type	Non-FDA Approved Drugs
Bladder Cancer	Sitravatinib (TKI), Adstiladrin (Gene Therapy), Bempegaldesleukin (Immune Modulation), TAR-200 (Intravesical Drug Delivery), Nadofaragene firadenovec
Brain Cancer	ONC201 (DRD2 Antagonist), SurVaxM (Peptide Vaccine), ABT-414 (EGFRvIII Target), PVSRIPO (Oncolytic Polio Virus), ACTR-T Cell Therapy
Breast Cancer	Camizestrant (SERD), Datopotamab deruxtecan (Trop-2 ADC), Zeno (HER2/HER3 Bispecific Antibody), MCLA-128 (HER2/3 Bispecific), Alpelisib/Letrozole
Cervical Cancer	CPI-613 (Metabolism Inhibitor), VB10.16 (HPV16 Vaccine), REGN2810 (PD-1 Antibody), Balstilimab (PD-1 Inhibitor)
Colorectal Cancer	Trilaciclib (CDK4/6 Inhibitor), DCC-3014 (CSF1R Inhibitor), ALX148 (CD47 Checkpoint Inhibitor), BBP-398 (SHP2 Inhibitor), MRTX849 (KRAS G12C Inhibitor)
Dermatofibrosarcoma Protuberans	PLX8394 (BRAF Inhibitor), Ripretinib (TKI targeting KIT, PDGFR)
Endocrine/Neuroendocrine Tumors	Surufatinib (VEGFR/FGFR Inhibitor), TTX-080 (CD39 Antibody), PRRT (Peptide Receptor Radionuclide Therapy), Evorpaccept (CD47-SIRPalpha Inhibitor)
Endometrial Cancer	Zilovetamab vedotin (ROR1 ADC), Mirvetuximab soravtansine (Folate Receptor Alpha ADC), MK-6482 (HIF-2alpha Inhibitor)
Esophageal Cancer	Tislelizumab (PD-1 Inhibitor), Sitravatinib (MET/TAM Receptor Inhibitor), Rovalpituzumab Tesirine (DLL3 ADC), Nivolumab + Relatlimab (LAG-3 inhibitor combo)
Head and Neck Cancer	Tipifarnib (Farnesyltransferase Inhibitor), Xevinapant (IAP Inhibitor), ADU-S100 (STING Pathway Activator), CUE-101 (HPV16 E7-targeted therapy)
Gastrointestinal Stromal Tumor	Ripretinib (KIT, PDGFR Inhibitor), DCC-2618 (Next-gen KIT inhibitor), Lenvatinib (Multi-TKI)
Giant Cell Tumor	Pexidartinib (CSF1R Inhibitor), AMG 820 (CSF1R Inhibitor)
Kidney Cancer	Savolitinib (MET Inhibitor), Belzutifan (HIF-2alpha Inhibitor), Nivolumab/Relatlimab (Checkpoint Combo), Cabozantinib (TKI)
Leukemia	Venetoclax (BCL-2 Inhibitor), Ivosidenib (IDH1 Inhibitor), Gilteritinib (FLT3 Inhibitor), Bomedemstat (LSD1 Inhibitor), CC-486 (Oral Azacitidine)
Liver/Bile Duct Cancer	Tremelimumab (CTLA-4 Inhibitor), Durvalumab + Tremelimumab (Checkpoint Combo), Ivosidenib (IDH1 Inhibitor), Cabozantinib (TKI), H3B-6527 (FGFR4 Inhibitor)
Lung Cancer	Sotorasib (KRAS G12C Inhibitor), Lazertinib (3rd-gen EGFR Inhibitor), Amivantamab (EGFR/MET Bispecific Antibody), DZD9008 (EGFR exon 20 Inhibitor), Lumakras (KRAS G12C)

Info/Databases used to evaluate each case

- Patient information – from Epic
- Biomarker testing result – most from Caris (integrated into Epic); some from Guardant, Tempus, FoundationOne, etc. (not integrated into Epic)
- OncoKB <https://www.oncokb.org/> to review specific biomarker/variant
- NCCN Guidelines
 - [Treatment guidelines](#) to review for therapy/timing of drug
 - [Genetic/Familial High-Risk Assessment](#) guidelines to review for germline recommendations
- [Lexidrug](#) to determine if drug is on-label or off-label
- [PubMed](#) to look for case reports, clinical trial results, etc.
- [ClinVar](#) to look up if variants have been described in germline (also provides some somatic info)
- [ClinicalTrials.gov](#) to look for clinical trials
- LabKey database to enter/house recommendations; review previous recommendations

Expert-Guided Large Language Models for Clinical Decision Support in Precision Oncology

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ABSTRACT

PURPOSE Rapidly expanding medical literature challenges oncologists seeking targeted cancer therapies. General-purpose large language models (LLMs) lack domain-specific knowledge, limiting their clinical utility. This study introduces the LLM system Medical Evidence Retrieval and Data Integration for Tailored Healthcare (MEREDITH), designed to support treatment recommendations in precision oncology. Built on Google’s Gemini Pro LLM, MEREDITH uses *retrieval-augmented generation* and *chain of thought*.

METHODS We evaluated MEREDITH on 10 publicly available fictional oncology cases with iterative feedback from a molecular tumor board (MTB) at a major German cancer center. Initially limited to PubMed-indexed literature (draft system), MEREDITH was enhanced to incorporate clinical studies on drug response within the specific tumor type, trial databases, drug approval status, and oncologic guidelines. The MTB provided a benchmark with manually curated treatment recommendations and assessed the clinical relevance of LLM-generated options (qualitative assessment). We measured semantic cosine similarity between LLM suggestions and clinician responses (quantitative assessment).

RESULTS MEREDITH identified a broader range of treatment options (median 4) compared with MTB experts (median 2). These options included therapies on the basis of preclinical data and combination treatments, expanding the treatment possibilities for consideration by the MTB. This broader approach was achieved by incorporating a curated medical data set that contextualized molecular targetability. Mirroring the approach MTB experts use to evaluate MTB cases improved the LLM’s ability to generate relevant suggestions. This is supported by high concordance between LLM suggestions and expert recommendations (94.7% for the enhanced system) and a significant increase in semantic similarity from the draft to the enhanced system (from 0.71 to 0.76, $P = .01$).

CONCLUSION Expert feedback and domain-specific data augment LLM performance. Future research should investigate responsible LLM integration into real-world clinical workflows.

ACCOMPANYING CONTENT

-  Appendix
-  Data Sharing Statement
-  Visual Abstract

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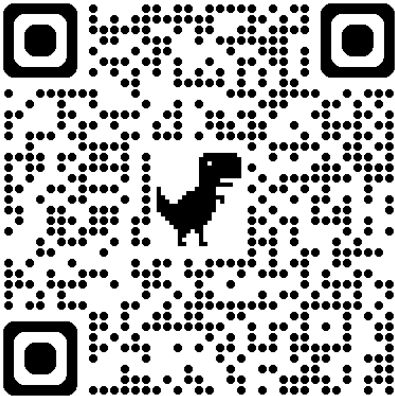
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CONTEXT

Key Objective
Can domain-specific large language models (LLMs) support personalized treatment recommendations within molecular tumor boards (MTBs)?

Knowledge Generated
Using a comprehensive data set of research studies, oncologic guidelines, clinical trials, and drug availability, an LLM system was configured with retrieval-augmented generation and chain of thought to generate targeted treatment options for 10 fictional patients with cancer. The analysis showed high concordance between LLM-generated and expert oncologist recommendations, indicating that replicating the precision oncologists’ evaluation process can enhance LLM performance for personalized cancer care.

Relevance
Domain-specific LLMs show potential for MTB decision support.





THANK YOU

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